

Andrew Yuan

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Technical Skills

Languages: Java, PHP, Python

Backend: MySQL, PostgreSQL, Redis, DynamoDB, Elasticsearch, RESTful APIs, Microservices, Spring Boot

Frontend: JavaScript, React, Vue

DevOps & Cloud: Azure, AWS, Docker, Kubernetes, CI/CD Pipelines

Testing: Agile/Kanban, Software Development Lifecycle, Performance Optimization, Communication Skills

Summary

Backend Software Engineer with 4+ years of experience designing and building scalable, secure, and DDD systems. Experienced in object-oriented programming, RESTful API design, and database-driven architectures. Experienced in Azure, AWS, SQL optimization, and high-concurrency services. Practical experience with AI-driven MCP (multi-channel processing) projects and AI agent development. Active open source contributor.

Experience

Uniuni Vancouver, BC
Software Developer July 2025- Nov 2025

- Contributed to a package sorting system handling over 500,000 orders daily, optimizing the order transit state process, and implementing a package tag system.
- Designed and implemented a scalable shipment tag management system using Laravel, leveraging event-driven architecture (Events/Listeners), distributed locking for concurrency control, repository pattern, and action-based tag.
- Optimized SQL query APIs by refactoring query logic and improving database indexing, reducing response time from approximately 1 minute to around 6 seconds.

TBoostAI Vancouver, BC
software engineer March 2025 - Aug 2025
Technology stack: **JavaScript, React, Java, Python, Milvus, Azure, RAG, LangChain**

- Participated in the development of an AI-powered automotive MCP, helping to recommend the most suitable used cars and dealers to users based on key profile information.
- Integrated the **Milvus** vector database using **Java** clients to enable semantic search and vector-based data handling, and utilized Cursor to improve coding efficiency.
- Designed and implemented a multi-stage state machine AI agent using **Springboot**. Integrated SMS (**Twilio**) and email sending functionalities.

China Unicom Xi'an Software Research Institute May 2021 - Dec 2023
Software Engineer
Tech Stack: Java, React, React Native, Redis, Kafka, Doris, Elasticsearch, AWS S3, Hibernate, MySQL, Docker, Kubernetes, DDD

- Led in the development of the 10010 Hotline product-recommendation module serving hundreds of millions of users: optimized **Redis** operations with **Lua**, utilized asynchronous and **strategy design pattern** to refactor the existing module, stress-tested to sustain **80,000 QPS** for this module, and boosted response speed by 5%. Reached the **TPS 5000**.
- Architected and optimized the China Unicom data visualization dashboard using **lazy rendering, front-end caching, SQL optimization, and result caching**, storing 20 GB of data on **Doris**, and improved query rendering time from 2 seconds to 200 ms.
- Build up and refactor the old questionnaire platform to conduct surveys for China Unicom users. Implemented real-time communication through sending SMS questionnaires, enabling users to interact via the Netty network framework. Integrated with an external call platform, allowing users to complete the questionnaire process via phone. Utilize MySQL's load data feature to import a lot of data.
- Developed the file system module that connects to Amazon Web Services (S3), ensuring Cloud Foundry standards of Production System deployments.
- Developed a work order system that receives data through Kafka and uses Elasticsearch to store log messages generated by the application, ensuring low latency in data retrieval and enhancing logistics processes.

ShunFeng Technology(Global 500)

Full Stack Engineer

Tech Stack: Java, VUE, Redis, Kafka, Echart, Caffeine

- Participated in a cash management system, enabling cash pooling, real-time account monitoring, and automated financial reporting to improve enterprise-wide fund visibility and efficiency.
- Java for performance-critical backend modules, with Lua used primarily in Redis scripting.
- Integrated a map drill-down feature using the **ECharts** framework to display financial data for each small region.
- Utilized **Vue** frameworks with core plugins such as **Vue-router**, **Vuex**, **Axios**, **element-ui**, **echarts**, and **vue_awesome** to build the cash management system front end.

May 2020 - May 2021

Education

New York Institute of Technology-Vancouver Cybersecurity	Master Degree • 3.94/4.0 Jan 2024 - Dec 2024
Washington State University Computer Science(Artificial Intelligent Track)	Master Degree Sep 2017 - Jan 2020
Jilin University of Finance and Economy Computer Science	Bachelor degree Sep 2013- Jul 2017